

# Exercise Class 1 — Consumer Theory (MWG Ch. 1–3)

Advanced Microeconomics 1 · Thursday 8 October, 11:00–13:00

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**Format (120 min).** 60 min: assigned students present Problems 1–5 at the board (assignments posted Monday 5 October). 30 min: instructor works the hard item (Problem 6). 30 min: timed prelim-style question (Problem 7), closed book, ungraded, self-marked against the sketch afterwards.

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## Part I — Board problems

### Problem 1 Rational preferences and their limits

- (a) Let  $\succsim$  be rational. Prove that  $\succ$  is transitive and irreflexive, and that  $\sim$  is an equivalence relation.
- (b) Give an example of a complete but intransitive relation on a three-element set, and exhibit an explicit money pump against an agent with those strict preferences.
- (c) Lexicographic preferences on  $\mathbb{R}_+^2$ :  $x \succsim y$  iff  $[x_1 > y_1]$  or  $[x_1 = y_1 \text{ and } x_2 \geq y_2]$ . Show they are rational and strongly monotone; show they are not continuous; sketch why no utility representation exists (map each  $x_1$  to a rational number in a suitably chosen interval and count).
- (d) For finite  $X$  and rational  $\succsim$ , verify that  $u(x) = \#\{y \in X : x \succsim y\}$  represents  $\succsim$ .

### Problem 2 Choice structures and WARP on a finite domain

Let  $X = \{a, b, c\}$  and  $\mathcal{B} = \{\{a, b\}, \{b, c\}, \{a, c\}\}$  (pairs only), with

$$C(\{a, b\}) = \{a\}, \quad C(\{b, c\}) = \{b\}, \quad C(\{a, c\}) = \{c\}.$$

- (a) Verify that this choice structure satisfies WARP.
- (b) Show that no rational preference rationalizes it.
- (c) Now add  $\{a, b, c\}$  to  $\mathcal{B}$ . Show that *no* specification of  $C(\{a, b, c\})$  preserves WARP. Reconcile with MWG Proposition 1.D.2.

### Problem 3 The CES workout

Let  $u(x_1, x_2) = (x_1^\rho + x_2^\rho)^{1/\rho}$ ,  $\rho < 1$ ,  $\rho \neq 0$ ; write  $\sigma = 1/(1 - \rho)$  and  $P = (p_1^{1-\sigma} + p_2^{1-\sigma})^{1/(1-\sigma)}$ .

- (a) Derive  $x(p, w)$  and  $v(p, w)$  from the UMP.

- (b) Derive  $h(p, u)$  and  $e(p, u)$  from the EMP.
- (c) Verify all four duality identities:  $e(p, v(p, w)) = w$ ;  $v(p, e(p, u)) = u$ ;  $h(p, u) = x(p, e(p, u))$ ;  $x(p, w) = h(p, v(p, w))$ .
- (d) Verify Roy's identity and Shephard's lemma by direct differentiation.
- (e) Recover Cobb–Douglas as  $\sigma \rightarrow 1$  and Leontief as  $\sigma \rightarrow 0$  (limits of  $e(p, u)$  suffice).
- (f) *The two accounting laws at work.* For a general demand system satisfying homogeneity of degree zero and Walras' law, differentiate the two identities to derive **Engel aggregation**  $\sum_{\ell} p_{\ell} \partial x_{\ell} / \partial w = 1$  and **Cournot aggregation**  $\sum_{\ell} p_{\ell} \partial x_{\ell} / \partial p_k = -x_k$ . Conclude that not every good can be inferior. Then verify both identities explicitly for the CES system of part (a).

### Problem 4 Quasilinear preferences: the welfare-exact case

Let  $u(x_0, x_1) = x_0 + \phi(x_1)$  with  $\phi' > 0 > \phi''$ ,  $p_0 = 1$ , and  $w$  large enough for interior solutions.

- (a) Derive  $x_1(p_1)$  (note: independent of  $w$ ),  $v(p_1, w)$ ,  $e(p_1, u)$ ,  $h_1(p_1)$ .
- (b) Show  $h_1(p_1, u) = x_1(p_1)$  for every  $u$ : Hicksian and Walrasian demand coincide.
- (c) For a price change  $p_1^0 \rightarrow p_1^1$ , prove  $CV = EV = \Delta CS = \int_{p_1^1}^{p_1^0} x_1(p) dp$ .
- (d) Exactly which step fails when wealth effects are restored? Point to the term in the Slutsky equation.

### Problem 5 Slutsky by hand, once in your life

For the CES demand system of Problem 3 with  $\sigma = 2$ ,  $p = (1, 1)$ ,  $w = 2$ :

- (a) Compute the  $2 \times 2$  Slutsky matrix  $S(p, w) = D_p x + D_w x x^{\top}$  numerically.
- (b) Verify symmetry, negative semidefiniteness (compute eigenvalues or check  $s_{11} \leq 0$  and  $\det S = 0$ ), and  $S p = 0$ .
- (c) Confirm  $S = D_p h$  evaluated at  $u = v(p, w)$ .
- (d) With  $L = 2$ , show from NSD + singularity alone that  $s_{12} \geq 0$ : two goods must be net substitutes.

## Part II — Instructor-worked hard item

### Problem 6 Afriat by hand

Consider  $T = 3$  observations on two goods:

| $t$ | $p_1^t$ | $p_2^t$ | $x_1^t$ | $x_2^t$ |
|-----|---------|---------|---------|---------|
| 1   | 2       | 1       | 3       | 2       |
| 2   | 1       | 2       | 2       | 3       |
| 3   | 1       | 1       | 4       | 1       |

- (a) Build the  $3 \times 3$  expenditure cross-matrix  $E_{ts} = p^t \cdot x^s$  and the direct revealed preference relations  $R^D$  (chosen bundle weakly cheaper comparison) and  $P^D$ .
- (b) Compute the transitive closure and check GARP.
- (c) If GARP holds, find Afriat numbers  $(U_t, \lambda_t)$  solving  $U_s \leq U_t + \lambda_t p^t(x^s - x^t)$  for all  $s, t$ , and write down the rationalizing utility  $u(x) = \min_t \{U_t + \lambda_t p^t(x - x^t)\}$ . If it fails, exhibit the violating cycle(s) and compute the CCEI by hand (find the budget scaling at which the last cycle just breaks).
- (d) Now replace the third observation by  $x^3 = (2, 2)$  (same prices). Re-check GARP; construct  $(U_t, \lambda_t)$  — try  $\lambda_t = 1$  — and the rationalizing  $u(x) = \min_t \{U_t + \lambda_t p^t(x - x^t)\}$ ; evaluate  $u$  at the three bundles and confirm each is optimal in its own budget set.

### Part III — Timed prelim question (30 minutes, closed book)

#### Problem 7 Reconstructing the square from $v$

A consumer's indirect utility is

$$v(p_1, p_2, w) = \frac{w}{p_1^\alpha p_2^{1-\alpha}}, \quad \alpha \in (0, 1).$$

- (a) Verify  $v$  has the properties required of an indirect utility function (h.d.0, monotonicity, quasi-convexity may be asserted with a one-line reason).
- (b) Recover Walrasian demand via Roy's identity.
- (c) Recover  $e(p, u)$  by inverting, and Hicksian demand via Shephard's lemma.
- (d) Compute the Slutsky matrix and verify symmetry, NSD, and  $Sp = 0$ .
- (e) Which familiar preferences generated this  $v$ ? Justify.

## Part IV — Take-home extension

### Problem 8 Cheapflation and the limits of the superlative index

*After Chen, Levell & O’Connell (2025), IFS WP 25/21, from Lecture 4. Not presented at the board; a full worked solution is in the instructor annex — self-mark, then bring questions to office hours.*

One physical good — bread — comes in two quality rungs: cheap (good 1) and premium (good 2). A household has Stone–Geary preferences

$$u(q_1, q_2) = (q_1 - \gamma)^\alpha q_2^{1-\alpha}, \quad \alpha = \frac{1}{2}, \quad \gamma = 1,$$

read as: one loaf of cheap bread per period is committed subsistence; taste is Cobb–Douglas above it. Initial prices  $p^0 = (1, 2)$ ; nominal spending is *fixed* at  $x = 4$  in both periods (stagnant incomes). The surge is cheapflation:  $p^1 = (1.35, 2.30)$  — the cheap rung +35%, the premium rung +15%.

- (a) For general  $\alpha \in (0, 1)$ ,  $\gamma \geq 0$ : derive Walrasian demand, the indirect utility function, and show

$$e(p, u) = \gamma p_1 + u \left( \frac{p_1}{\alpha} \right)^\alpha \left( \frac{p_2}{1 - \alpha} \right)^{1-\alpha}.$$

For which parameter value are these preferences homothetic?

- (b) Compute the period-0 bundle,  $u^0$ , and the budget share of cheap bread.
- (c) Compute the Laspeyres index  $P_L$  and the Konüs index at initial living standards,  $P_K(u^0)$ . Verify  $P_L \geq P_K(u^0)$  and interpret the gap.
- (d) Compute the period-1 bundle and  $u^1$ . Show the cheap-bread budget share *rises* even though the cheap rung’s relative price rose. Decompose the share change using the compensated (Hicksian) share at  $(p^1, u^0)$ : how much is substitution, how much is trading down (the income effect)?
- (e) Compute the Paasche index  $P_P$ , the Fisher index  $P_F = \sqrt{P_L P_P}$ , and  $P_K(u^1)$ . Establish the full ordering of all five numbers. Which of the classical index-number inequalities survive non-homotheticity, and why does the “superlative” index approximate *neither* true cost-of-living index here?
- (f) Now a richer household: same preferences,  $x = 10$  in both periods. Compute its  $P_K(u^0)$  and the poor–rich gap in true inflation. Relate this to Chen–Levell–O’Connell’s 7.4pp finding, and state — with a defence in terms of CV or EV — which of part (e)’s five numbers automatic indexation *should* pay this household.

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